

Amelia Gibbs

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Education

Trinity University
B.A. in Mathematics

San Antonio, TX
2021 – 2025

University of California - San Diego
Ph.D. in Mathematics

San Diego, CA
2025 – Present

Research Experience

Trinity University

San Antonio, TX

Research Assistant; Advisor: Hoa Nguyen

Summer 2023

Optimized the method of images for regularized Stokeslets using theoretical and experimental results of spherical motion near a boundary

- Simulated spherical motion near a boundary at low Reynolds number with the method of images for regularized Stokeslets in MATLAB
- Developed approximations for optimal numerical parameters of various regularization functions from simulation data

University of Michigan - Dearborn

Dearborn, MI

Research Assistant; Advisor: Kelly Jabbusch

Summer 2024

Worked jointly with three other undergraduates to study error-correcting codes from toric varieties and geometric invariants of lattice polytopes

- Developed a new method for computing the minimum distance of toric codes associated to certain classes of lattice polygons
- Proved results regarding the affects of polytope operations on the Minkowski Length, a geometric invariant, of a lattice polytope

Publications

- Hoa Nguyen, Amelia Gibbs, Frank Healy, Orrin Shindell, Ricardo Cortez, Kate Brown, Jonathan McCoy, and Bruce Rodenborn. “*Optimizing the Method of Images for Regularized Stokeslets Using Theory and Experiments of Spheres Moving near Boundaries.*” *Physical Review Fluids* 10(3):033101, March 15 2025. doi:10.1103/PhysRevFluids.10.033101.
- Amelia Gibbs, and Brian K. Miceli. “*Two Combinatorial Interpretations of Rascal Numbers.*” *Journal of Integer Sequences* 27(8):21, November 12 2024. Article 24.8.2.
- Amelia Gibbs, Eliza Hogan, Kelly Jabbusch, Jenna Plute, and Nicholas Toloczko. “*Toric Surface Codes and the Periodicity of Polytopes.*” In preparation.

Presentations

- “*Calibration of the method of images for regularized Stokeslets using sphere motion near a boundary*,” APS Division of Fluid Dynamics 76th Annual Meeting, November 2023, Washington DC, Poster Presentation.
- “*Two Combinatorial Interpretations of Rascal Numbers*,” CombinaTexas, March 2024, Texas A&M University, Contributed Talk.
- “*Toric Surface Codes and the Periodicity of Polytopes*,” SUMMR, July 2024, Michigan State University, Contributed Talk, Presented Jointly.
- “*Toric Surface Codes and the Periodicity of Polytopes*,” Young Mathematicians Conference, August 2024, Ohio State University, Contributed Talk, Presented Jointly.
- “*A Combinatorial Proof of Schläfli’s Identity and Gould’s Identity*,” AMS Fall 2024 Central Sectional Meeting, September 2024, University of Texas - San Antonio, Invited Talk.
- “*Toric Surface Codes and the Periodicity of Polytopes*,” Joint Mathematics Meetings, January 2025, Seattle WA, Contributed Talk, Presented Jointly.
- “*A Combinatorial Proof of Schläfli’s Identity and Gould’s Identity*,” CombinaTexas, April 2025, Texas A&M University, Contributed Talk.

Awards

Junior Mathematics Award

2024

Given to the junior whose performance and promise in mathematics are deemed the best by the mathematics faculty at Trinity

Teaching Experience

Trinity University

San Antonio, TX

Teaching Assistant

2023 – 2025

- TA for Linear Algebra (Math 3323) Spring 2025
- TA for Combinatorics (Math 3343) Fall 2024
- TA for Modern Algebra (Math 3362) Spring 2024
- TA for Applied Linear Algebra (Math 3319) Fall 2023
- TA for Linear Algebra (Math 3323) Spring 2023

References

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Hoa Nguyen, PhD
Professor
Department of Mathematics
Trinity University
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Kelly Jabbusch, PhD
Associate Professor
Department of Mathematics and Statistics
University of Michigan
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